Let p be a prime number.

Lemma. $\Phi_p(x) = \frac{x^p - 1}{x-1} = x^{p-1} + x^{p-2} + \cdots + x + 1$

is irreducible.

proof. observe that $\Phi_p(x+1) = [(x+1)^p - 1]/x = \frac{1}{x}\cdot\left(x^p + \binom{p}{1}x^{p-1} + \cdots + \binom{p}{p-1}x + \binom{p}{0} - 1\right)$

$$= x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + p,$$

so applying the eisenstein Criterion, we obtain the result.

Lemma. Let $\zeta_p \overset{\text{def}}{=} \exp(2\pi i \cdot \frac{1}{p})$. Then, the set

$$\{\zeta_p^0, \zeta_p^1, \cdots, \zeta_p^{p-1}\} \cong \mathbb{Z}_p.$$

Moreover, $\mathbb{Q}(\zeta_p)$ is the splitting field of $x^p - 1$ and also Φ_p, with $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1$.

proof. It is trivial, since p is a prime number. ☒

Lemma. $[\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}] = \frac{p-1}{2}$.

Proof. since $\zeta_p^{-1} = \zeta_p^{p-1} = \exp(2\pi i \cdot \frac{p-1}{p})$, $\zeta_p + \zeta_p^{-1} = 2\cdot\cos(2\pi \cdot \frac{1}{p}) \in \mathbb{R}$

Then, $x^2 - (\zeta_p + \zeta_p^{-1})x + 1 \in \mathbb{Q}(\zeta_p + \zeta_p^{-1})[x]$ and it has roots as ζ_p and ζ_p^{-1}.

For odd prime p ($p \geq 3$), $\zeta_p \notin \mathbb{Q}(\zeta_p + \zeta_p^{-1})$, so the polynomial above is irreducible.

Therefore, $[\mathbb{Q}(\zeta_p) : \mathbb{Q}(\zeta_p + \zeta_p^{-1})] = 2$. Now, the tower law gives the result.

(If $p = 2$, then $\zeta_2 = -1$, so proved.)

Thm. splitting field of $x^p - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$, and $[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}] = p(p-1)$.

proof. Since for each $0 \leq i \leq p-1$, $(\sqrt[p]{2}\cdot\zeta_p^i)^p = 2\cdot\zeta_p^{pi} = 2\cdot 1 = 2$,

so the splitting field of $x^p - 2$ is $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$. To evaluate the degree, observe that

$$\mathbb{Q}(\sqrt[p]{2}, \zeta_p) = \mathbb{Q}(\sqrt[p]{2})\mathbb{Q}(\zeta_p)$$ and

$[\mathbb{Q}(\sqrt[p]{2}) : \mathbb{Q}] = p$ and $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1$, so

$$[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}] = p(p-1).$$

Example. $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$

The K is a splitting field for $(x^2-2)\cdot(x^2-3)$

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt{2},\sqrt{3}) \arrow[dl, dash, "2"'] \arrow[d, dash, "2"'] \arrow[dr, dash, "2"] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash, "2"'] & \mathbb{Q}(\sqrt{6}) \arrow[d, dash, "2"'] & \mathbb{Q}(\sqrt{3}) \arrow[dl, dash, "2"] \\
 & \mathbb{Q} &
\end{tikzcd}$$

But, the splitting field of x^3-2 is not just $\mathbb{Q}(\sqrt[3]{2})$,
because there are three roots;

$$\sqrt[3]{2}, \quad \sqrt[3]{2}\cdot\left(\frac{-1+i\cdot\sqrt{3}}{2}\right), \quad \sqrt[3]{2}\cdot\left(\frac{-1-i\sqrt{3}}{2}\right)$$

Let K be the splitting field of x^3-2. That is,

$$\sqrt[3]{2} \in K \quad \text{and} \quad \sqrt[3]{2}\cdot\left(\frac{-1+i\sqrt{3}}{2}\right) \in K,$$

So $\sqrt{-3} \in K$. It follows that $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$.

Since $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) = \mathbb{Q}(\sqrt[3]{2})\,\mathbb{Q}(\sqrt{-3})$ and

$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ and $[\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] = 2$, these are relatively prime,
thus the degree of Composite field is 6.

Find the Galois group of $x^5-3 \in \mathbb{Q}(\zeta)[x]$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity.

Proof. Let K be the splitting field of x^5-3.
Since all roots of x^5-3 are

$$\sqrt[5]{3}\cdot\zeta^{k} \quad (k=0,1,2,3,4) \quad \text{where } \zeta = \exp\left(2\pi i\cdot\tfrac{1}{5}\right) = e^{\frac{2}{5}\pi i},$$

Therefore $K=\mathbb{Q}(\sqrt[5]{3},\zeta)$. Now,

$$\begin{tikzcd}
 & K=\mathbb{Q}(\sqrt[5]{3},\zeta) \arrow[dl, dash, "4"'] \arrow[dr, dash, "5"] & \\
\mathbb{Q}(\sqrt[5]{3}) \arrow[dr, dash, "5"'] & & \mathbb{Q}(\zeta) \arrow[dl, dash, "4"] \\
 & \mathbb{Q} &
\end{tikzcd}$$

$\left(\because x^5-3 \text{ is irreducible over } \mathbb{Q}\right)$

$\left(x^4+x^3+x^2+x+1 \text{ has root as } \zeta \text{ and irreducible}\right)$

Since $\mathbb{Q}(\sqrt[5]{3},\zeta)=\mathbb{Q}(\sqrt[5]{3})\cdot\mathbb{Q}(\zeta)$ and both degree over $\mathbb{Q}$ are relatively prime, $[K:\mathbb{Q}]=20$. And, since $[K:\mathbb{Q}(\zeta)]=5$, prime,

$$\mathrm{Gal}(K/\mathbb{Q}(\zeta)) \cong \mathbb{Z}_5.$$